Anand Mohan Pandey

+91-7463969003
Embedded System | Firmware anandmohan.amp@gmail.com | LinkedIn | GitHub

Professional Experience

VyomOS Jun 2025 - Present
Embedded Software Engineer
- Joined full-time after successful delivery of Netrajaal (A Mesh connectivity through LoRa with edge cam computing for surveillance in network prone area) to productize the prototype and scale it for field deployment.
- Worked on successful delivery of Micropython driver for SX1262 and contributed to the robotics layer for mesh connectivity and short path for data transfer.
- Successfully delivered and deployed these devices for surveillance in Network prone areas.
- **Teck-stack-** MicroPython, async, C/C++, makefile, cross-tool chain and build systems

Freelance Embedded Systems Developer Jan 2025 - May 2025
Clients: Cognecto
- Contributed to fix bugs in GPS tracker firmware with OTA, Geofencing, and optimized MQTT
- Implemented CAN J1939 protocol stack (SAE-compliant) for heavy machinery telemetry.

Clients: Aura Aerospace
- Built a MAVLink C++ library for Raspberry Pi Pico and a companion computer prototype.

Clients: VyomOS
- Netrajaal (Defence Surveillance System for network prone area): Designed and developed a prototype with SX1262 LoRa, OpenMV for Proof of Concept for mesh connectivity and image transfer through LoRa for long Range Transmission with RSA-based encryption and security mechanisms.

Okulo Aerospace July 24 - Jan 25
Embedded Software Engineer (Full-time)
- Developed a data gathering device to improve the aerodynamics of Fixed wing UAV.
- **Tech-Stack :** C/C++ | Ardupilot | STM32CubeIDE | ROS2 | STL

UrbanMatrix Technologies Aug 23 - June 24
Embedded Software Engineer (Full-time)
- Contributed to a companion computer system leveraging MAVSDK and WebSocket++ libraries to enable real-time telemetry streaming and command execution with minimal latency.
- Integrated SATCOM module with UART and implemented AT command Sequence with timing.
- Developed firmware for the controller firmware stack to add a Hardware Tampering system in our drones.
- **Tech Stack :** C | UART | I2C | Pixhawk | Control Algoritham | STM32CubeIDE | AWS-iot | ROS2

InstatantPost (NextGen Consumer Electronics Ecosystem for Smart Homes) May 22 - Feb 23
Power Electronics Design (Internship)
- Write firmware for microcontrollers to control and monitor power conversion systems, Perform board bring-up, including initial power-on tests and basic functionality checks, Assist in PCB layout and design for prototype boards using KiCAD.
- **Tech Stack :** C | Circuit Design | KiCad | Esp32 | oscilloscope

Technical Skills

Programming, OS & Framework : C/C++, Micro-Python, Github, makefiles, cross-toolchains, and build systems, FreeRTOS, Linux

Debugging : SWD, Logic Analyzer, JTAG

Education

Degree	Institute/Board Stream	CGPA/Percentage	Year
Bachelor in Engineering	Electrical & Electronics Engineering (DSCE)	7.6	2019-2023
Senior Secondary	Shri Chaitnya (CBSE)	78	2018
Secondary	Sainik School Ambikapur (CBSE)	9.4	2016

Projects

SX1262 LoRa Communication Driver [Link]
- Developed a MicroPython driver library for the Semtech SX1262 LoRa transceiver enabling SPI-based long-range wireless communication (LoRa & FSK support) for embedded Python devices.
- Implemented core radio functions (initialization, send/receive, config) and example usage to demonstrate reliable LoRa packet transmission over sub-GHz links.

PID
Self-Directed Project [Link]
- A simple PID (Proportional-Integral-Derivative) controller in C, basic simulation of a temperature control system to understand the principles of PID control and its application in process control systems.
- **Tech-Stack/concept :** C | PID Control | Discrete-time implementation | Makefile

Adaptive RTOS Task Scheduler for Multi-Sensor Data Fusion
Self-Directed Project [Link]
- Engineered an adaptive RTOS-based task scheduler optimising CPU usage for multi-sensor data fusion
- Implemented dynamic priority adjustment and flexible sensor integration using advanced FreeRTOS features.
- Developed real-time data fusion algorithms for efficient processing of multi-source sensor data.
- **Tech-Stack/concept :** C | FreeRTOS | CMake | git

Publication

Journal : Suganthi Neelagiri, Ritesh Kumar, Rishabh Dadhich, Yaman Singh, Anand Mohan Pandey, **“Hexapod Robot Platforms: Applications in Real-world Environment”**. [Link]

Extra-Curricular Achievements

People 10 Code Combat: Third Position, Indoor navigation system, Designed Electronic subsystem.
PRODUCT-A-THON: Winner DSCE hackathon 2022 (annual hackathon of DSCE).
Seminar Competition DSCE: Winner, 2020. Runner up, 2019. Spoke about memristors and nanotechnology

Key Courses Taken

EE: Embedded Systems, Digital Circuits, Circuit Theory, Control System, Signal and Systems, Basic Electronics, Power Electronics.
Math & Others: Analysis and Design of Algorithm, Linear Algebra, Device Driver Development, Networking, LoRA communication, I2C, SPI, UART, Flash